Statistical Pileup Correction Method for HPGe detectors

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Abstract

We consider a nonlinear inversion problem occurring in gamma spectrometry. In that framework, photon energies are converted to electrical pulses which are susceptible to overlap, creating clusters of pulses, referred to as *pileup*. This phenomenon introduces a distortion that can be a nuisance for the correct identification of the radionuclides. In that application we are interested in the distribution of the individual photon energies, hence it is necessary to estimate this distribution using the indirect observations of the piled-up electrical pulses. In this article we present a new method to correct the pileup phenomenon, based on a nonlinear formula, which is inverted to give a nonparametric estimator of the individual energy distribution. Since the proposed estimator depends on parameters to be chosen carefully by the user, we also detail data-driven selection methods for the associated algorithm. Applications on simulations and real datasets are presented, which shows the efficiency of the proposed method.

Index Terms

nonparametric density estimation, density deconvolution, gamma spectrometry, HPGe detectors, non-linear inverse problems.

I. Introduction

One of the aims of nuclear spectrometry is to analyze the energy distribution of photons (X or Gamma) emitted by a radioactive source. The usual technique consists in converting the photon energy into current in a semiconductor detector. The photons impinging on the detector create pairs of electron-holes, the number of which being proportional to the energy deposited in the semiconductor. These electron-hole pairs lead to charge carriers with opposite polarity, whose migration creates an electrical current pulse which is measured. Under appropriate experimental conditions (low temperature, high purity crystal), the integral of this pulse of current over time is proportional to the total amount of charge carriers, thus giving means to measure the energy deposited; for an exhaustive description of the photon-matter interactions that occur in the detector, the reader is referred to [1] and [2].

In many practical experiments, the successive times of photon-matter interactions may be modelled as a homogeneous Poisson process. The corresponding inter-arrival times may be shorter than the mean duration of generated electrical pulses after amplification, thus creating *clusters* of pulses (see Figure 1). In gamma ray spectrometry, this phenomenon is referred to as *pileup*. The distribution of the energy deposited in the detector is usually estimated by means of a histogram of the collected energies. This histogram, in nuclear science, is referred to as *energy spectrum*. The pileup phenomenon causes a distortion of the energy spectrum which becomes more severe as the incoming counting rate increases. In particular it can create fake peaks and a distortion of the Compton continuum (that is the continuous part of the histogram related to photons whose energy is partially recorded, see [1]), which both can cripple the identification of the radionuclides (see Figure 2). More precisely, the photons corresponding to Compton continuum can also pileup, introducing a continuum at higher energies, which can mask some peaks of interest at high energy whose occurrence rate is low. Since clusters may contain several pulses, the pileup phenomenon also introduces a negative bias in the naive estimation of the intensity of the underlying Poisson process (that is, counting all events and use the Poisson assumption without taking the hidden counts into consideration), and thus also in the measurement of the lifetime of the radioactive elements. This problem has been studied by many authors in the field of nuclear instrumentation since the 1960's (see [3] for a detailed review of these early contributions; classical pileup correction techniques are detailed in the ANSI norm [4]).

In mathematical terms, the pileup problem can be formalized as follows. Denote by $\{T_k,\, k \geq 1\}$ the sequence of arrival times of the photons, assumed to be the ordered points of a homogeneous Poisson process. The current intensity as a function of time can be modelled as a shot-noise process

$$W(t) \stackrel{\text{def}}{=} \sum_{k \geq 1} F_k(t - T_k) \;, \tag{1}$$

where $\{F_k(s),\, k \geq 1\}$ are the contributions of each individual photon to the overall current intensity. By analogy with queuing models, we call $\{W(t),\, t \geq 0\}$ the *workload* process. The current pulses $\{F_k(s),\, k \geq 1\}$ are assumed to be independent copies of a continuous time stochastic process $\{F(s),\, s \geq 0\}$ whose path has a finite support $[0, X]$ a.s., where X is the pulse duration (the duration of the collection of charge carriers). The integral of the pulse $Y \stackrel{\text{def}}{=} \int_0^X F(s)\, ds$ is equal to the energy of a single photon. The restriction of the workload process to a maximal segment where it is positive (resp. 0) is referred to as a *busy* (resp. *idle*) *period*. An idle period followed by a busy period is called a *cycle*.

In our experimental setting, the sequence of pulse duration and energy $\{(X_k, Y_k),\, k \geq 1\}$ is not directly observed. Instead, the only available data are the durations of the busy and idle periods and the total

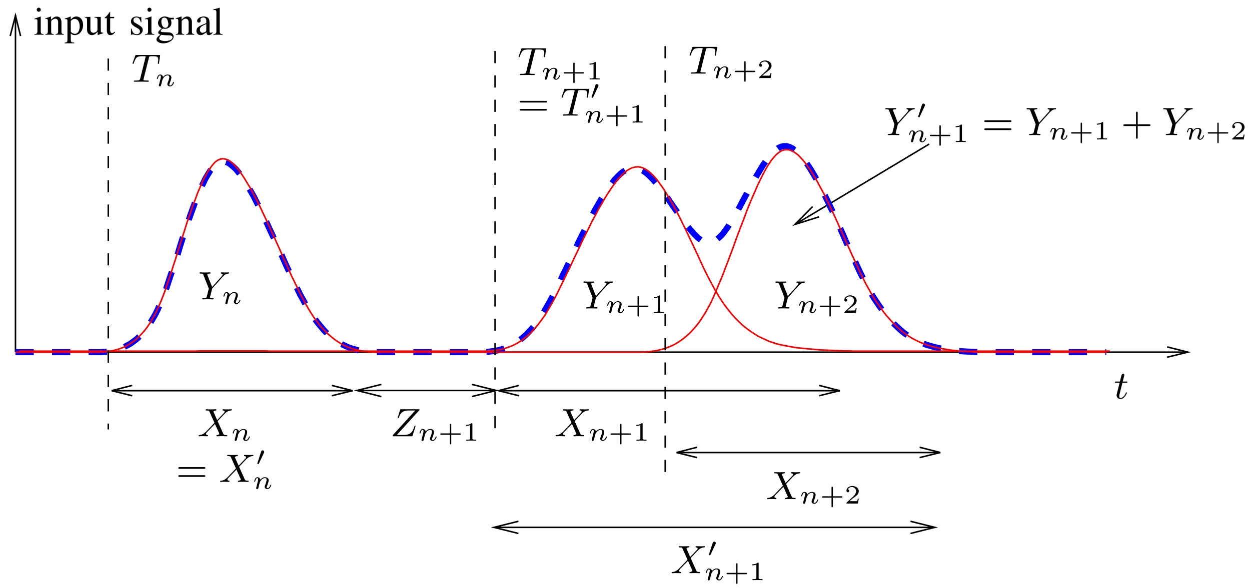


Fig. 1. Illustration of Type II Counter Problem: input signal with arrival times T_k, lengths X_k and energies Y_k, $k = n, \ldots, n+2$ associated observations times $T'_{k'}$, idle period durations $Z_{k'}$ busy period durations $X'_{k'}$ and energies $Y'_{k'}$, $k' = n, n+1$.

amounts of charge carriers collected during busy periods. Define $\{T'_k,\, k \geq 1\}$ the ordered sequence of busy period arrivals and $\{X'_k,\, k \geq 1\}$ the corresponding sequence of durations. We further define, for all $k \geq 1$, $Y'_k \stackrel{\text{def}}{=} \int_{T'_k}^{T'_k+X'_k} W(t)\, dt$, the total amount of charge of the k-th busy period. Finally we denote by Z_k the duration of the k-th idle period, that is $Z_1 = T_1$ and, for all $k \geq 2$, $Z_k \stackrel{\text{def}}{=} T'_k - (T'_{k-1} + X'_{k-1})$. Thus, the problem of pileup correction can be seen formally as the estimation of the probability density function (pdf) m (referred to as the *energy spectrum* in the spectrometry literature) of the photon energy Y, having observed n cycles $\{(Z_k, X'_k, Y'_k),\, 1 \leq k \leq n\}$.

Early papers on pileup correction focus specifically on activity correction methods, such as the VPG (Virtual Pulse Generator) method described in [5] and [6]. In these contributions, the pileup is considered as an artifact to be removed, for example using an inhibitor circuit (see e.g. [7]); another way to correct pileup is also described in [8], where the measured pulse of current is correlated with a theoretical pulse shape, thus allowing to do inference on the joint distribution. However, as detailed in [9], if the activity correction has been extensively studied, the pileup phenomenon has received up to our knowledge little attention, despite its importance at high counting rates. Moreover, it must be stressed that these techniques are strongly related to the instrumentation used for the experiments.

These considerations, as well as theoretical contributions of [10], [11], [12], show the necessity to use mathematical models taking into account the stochastic aspects of the input signal and lead to methods

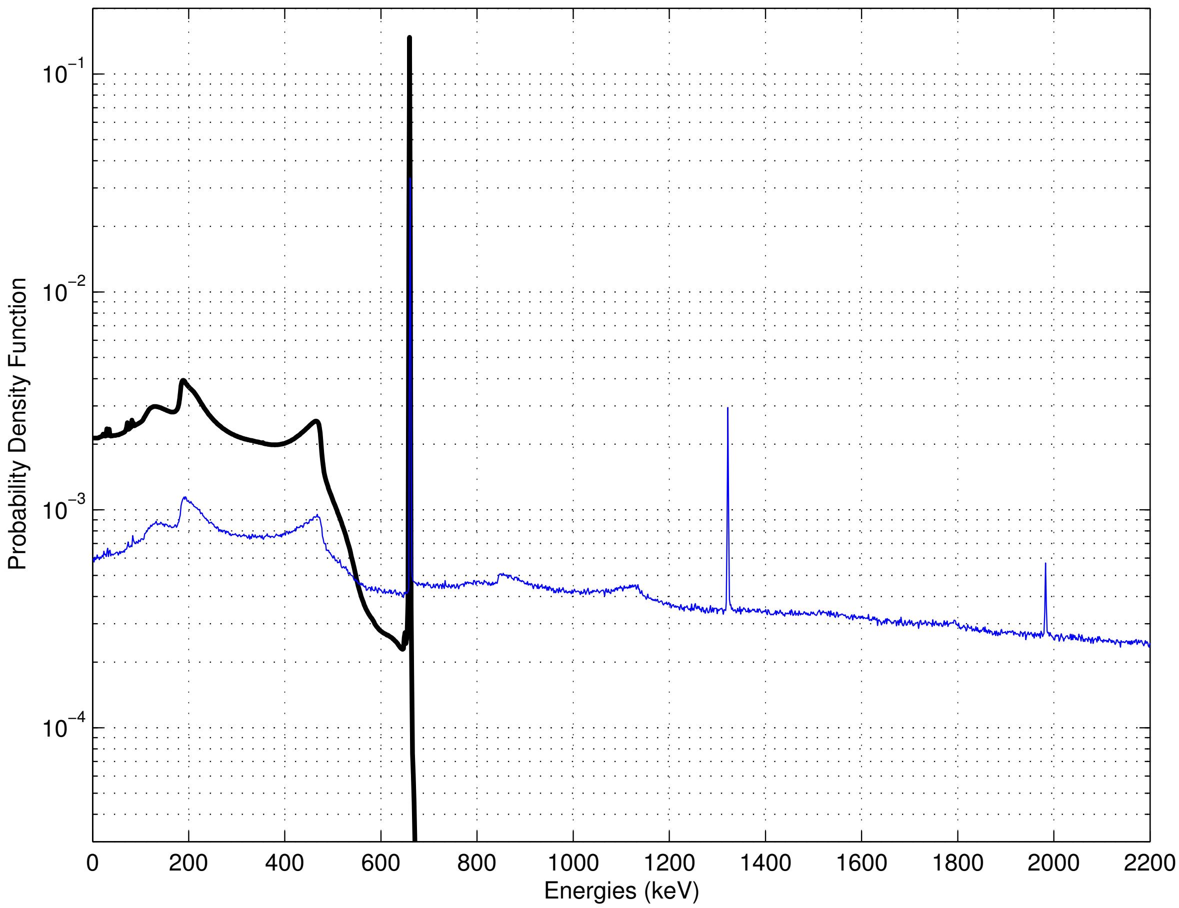


Fig. 2. Theoretical probability density function of the photon-matter interaction energy for the Cesium 137 (black line) and observed piled-up histogram at a counting rate $\lambda = 1000000$ photons per second (blue line).

for pileup correction instead of rejection, as detailed in [13] and more recently in [14]. Still, the methods described in these articles do not focus on the pileup phenomenon, but rather on the estimation of the intensity of the Poisson process. Instead of a rejection method, we propose in this contribution to tackle the inverse problem of recovering the pulses energy distribution from the cluster durations and energies observations. This presents some interesting advantages: it is unnecessary to make any prior on the distribution of the lengths of the pulses, allowing the method to be adaptable to any kind of instrumentation; another advantage is that they can be used off-line to process previously acquired data, or in complement of another online method.

This problem also shares similarity with the inference of the service time in M/G/∞ (see [15]) queue from the duration of the busy periods. In that framework, the arrival times of jobs at a server are taken to be the points of a Poisson process, there is an infinity of servers and service times have an essentially arbitrary distribution of interest. Based on the durations of the busy periods X' and of inter-arrival times between the busy periods Z, we want to estimate the probability density of service time X. We can find in [16] and more recently in [17] a relation between the cumulative distribution functions (cdf) of X and X' and [18] derived from this formula an estimator of the pdf of X for the study of queueing traffic. Our approach is motivated by the physical observation that pulse durations and energies are not independent. Though quantum aspects such as Compton effect are sufficient to explain the randomness of the energy distribution, the duration and the energy of detected pulses both depend on other parameters, for example the shape of the detector or the place of interaction, which are difficult to exploit to study the pileup phenomenon. With these previous considerations, the proposed approach in this article is to develop a nonparametric pileup correction (instead of rejection) method, based on the methodology developed in [19], and to focus on the study of the numerical implementation, particularly on the choice of the parameters.

The paper is organized as follows. We introduce in Section II an estimator used for pileup correction, and the underlying algorithm. This estimator differs from the one presented in [19] by its pre-regularization. Section III considers the problem of the data-driven choice of the parameters of the estimator, which is of importance when the number n of samples $\{(X'_k, Y'_k),\ k = 1 \dots n\}$ is fixed. We present in Section IV results on simulations concerning the selection methods described in Section III. Finally, some results on real data are also discussed in Section V.

II. Description of the pileup correction algorithm

In this section we recall the main theorem introduced for the pileup correction of nuclear spectroscopy, as well as the estimator obtained from this theorem. Detailed proofs of different assertions can be found in [19] or [20] and are omitted here for brevity.

A. *Main assumptions*

All along the paper, we suppose that

(H-1) $\{T_k,\ k \geq 1\}$ is the ordered sequence of the points of a homogeneous Poisson process on the positive half-line with intensity λ.

(H-2) $\{(X, Y), (X_k, Y_k),\ k \geq 1\}$ is a sequence of independent and identically distributed (*i.i.d.*) $(0, \infty)^2$-valued random variables with probability distribution denoted by P and independent of $\{T_k,\, k \geq 1\}$. In addition, Y has a marginal pdf $m(y)$ on $y \in \mathbb{R}_+$ and the expectations $\mathbb{E}[X]$ and $\mathbb{E}[Y]$ are finite.

Formally, $\{(T_k, X_k, Y_k),\, k \geq 1\}$ is a Poisson point process with control measure $\lambda\mathrm{Leb} \otimes P$, where Leb denotes the Lebesgue measure on the positive half-line. Let us recall a few basic properties satisfied under this assumption by the sequence $\{(Z_k, X'_k, Y'_k),\, k \geq 1\}$ defined in the introduction. Using standard properties of Poisson point processes (see e.g. [20]), it can be shown that

(i) $\{(Z_k, X'_k, Y'_k),\, k \geq 1\}$ is a sequence of *i.i.d.* random triples,

(ii) $\{Z_k,\, k \geq 1\}$ and $\{(X'_k, Y'_k),\, k \geq 1\}$ are two independent sequences,

(iii) Z_k has exponential distribution with parameter λ for all k.

We denote by (X', Y') a couple having the same distribution as the variables (X'_k, Y'_k), $k \geq 1$ and by P' its probability measure. Let $\mathbb{C}_+$be the closed half-plane of complex numbers with nonnegative real parts and $\mathbb{C}_+^*$be the open half-plane of complex numbers with positive real parts. The following formula is stated in [21], and its proof is detailed in [19] (see also [20]). For all $(s,p) \in \mathbb{C}_+^* \times \mathbb{C}_+$,

$$\int_0^{+\infty} \mathrm{e}^{-(s+\lambda)x} \left[\exp\left(\lambda \int_0^{\infty} \mathrm{e}^{-py} k(x, dy)\right) - 1\right] dx = \frac{\lambda \mathcal{L}P'(s,p)}{s+\lambda} \frac{1}{s + \lambda - \lambda\mathcal{L}P'(s,p)}\ , \quad (2)$$

where, for any $x \geq 0$,

$$k(x, \cdot) \overset{\text{def}}{=} \int_0^x (x-u) P(du, \cdot) \quad (3)$$

and $\mathcal{L}P'$ is the Laplace transform of P',

$$\mathcal{L}P'(s,p) = \iint_{\mathbb{R}_+ \times \mathbb{R}_+} \mathrm{e}^{-su} \mathrm{e}^{-pv} P'(du, dv)\ .$$

Eq. (2) provides a relation between λ, the joint distributions of the energy and duration of a busy period P' and of a single pulse P. Our objective is the nonparametric estimation of the pdf m of Y, that is the marginal distribution along the second coordinate of P, having observed $\{(Z_k, X'_k, Y'_k),\, k = 1, \ldots, n\}$. We present in the next section the inverse problem methodology that we use to derive an estimator of m based on the relation (2).

B. An inversion formula

By the Radon-Nikodym Theorem (see [22]), the existence of a pdf for Y implies that, for any $x > 0$, the measure $A \mapsto P([0,x] \times A)$ defined on Borel sets also has a density function (although not a probability density function) on $\mathbb{R}_+$, which we denote by $m_x(y)$, $y \in \mathbb{R}_+$. By definition of m_x, it is easily seen that $\|m - m_x\|_1 \leq \mathbb{P}(X > x) \to 0$ as $x \to \infty$ and that the sequence of densities $\{m_x,\ x > 0\}$ is increasing: for all positive numbers $x \leq x'$ and for almost every $y \in \mathbb{R}$ (*w.r.t.*the Lebesgue measure), $m_x(y) \leq m_{x'}(y) \leq m(y)$. Hence estimating the density m_x amounts to estimate m with a negative bias which vanishes as x becomes *large*.

By inverting (2) we want to relate m with P' and λ, which will be easily estimated using the *i.i.d.* sample $\{(Z_k, X'_k, Y'_k),\ k = 1, \ldots, n\}$. In fact m is not directly involved in (2) but m_x is implicitly involved through the definition of $k(x, \cdot)$ in (3) as shown below. In order to derive an inversion formula for $m_x(y)$ for all y, we further need some basic smoothness condition on this density, usually expressed in terms of Sobolev spaces, that is $m \in \mathcal{W}(\beta)$ where

$$\mathcal{W}(\beta) \stackrel{\text{def}}{=} \left\{ g \in L^2(\mathbb{R})\ ;\ \int_{-\infty}^{\infty} (1 + |\nu|)^{2\beta} |g^*(\nu)|^2\, d\nu < \infty \right\}\ ,$$

where β is the exponent of the Sobolev space in consideration and $g^*(\nu) = \int_{\mathbb{R}} g(y)\, \mathrm{e}^{-\mathrm{i}\nu y}\, dy$ denotes the Fourier transform of g.

Let us denote by Ψ the complex-valued function that appears in the right hand side (RHS) of (2), defined for all (ω, p) in $\mathbb{R} \times \mathbb{C}_+$ as

$$\Psi(\omega, p) \stackrel{\text{def}}{=} \frac{\lambda \mathcal{L}P'(c + \mathrm{i}\omega, p)}{(c + \mathrm{i}\omega + \lambda)(c + \mathrm{i}\omega + \lambda - \lambda \mathcal{L}P'(c + \mathrm{i}\omega, p))}\ , \quad (4)$$

where c is an arbitrary positive constant which makes the function $\omega \mapsto \Psi(\omega, p)$ integrable. Therefore, we may define, for all (x, p) in $\mathbb{R}_+^* \times \mathbb{C}_+$,

$$a(x, p) \stackrel{\text{def}}{=} 1 + \frac{\mathrm{e}^{(c+\lambda)x}}{2\pi} \int_{-\infty}^{+\infty} \Psi(\omega, p) \mathrm{e}^{\mathrm{i}\omega x}\, d\omega\ ; \quad (5)$$

$$I_1(x, p) \stackrel{\text{def}}{=} \iint_{\mathbb{R}_+^2} \mathbb{1}_{\{u \leq x\}} \mathrm{e}^{\lambda u - pv} P'(du, dv)\ ; \quad (6)$$

$$I_2(x, p) \stackrel{\text{def}}{=} \frac{\mathrm{e}^{(c+\lambda)x}}{2\pi} \times \int_{-\infty}^{+\infty} \mathcal{L}P'(c + \mathrm{i}\omega, p) \Psi(\omega, p) \mathrm{e}^{\mathrm{i}\omega x}\, d\omega\ . \quad (7)$$

The following result then provides the announced inversion formula for m_x.

Proposition 2.1: Let $x > 0$ and assume that $m_x \in \mathcal{W}(\beta)$ with $\beta > 1/2$. Then for all $y \in \mathbb{R}_+$,

$$m_x(y) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \frac{I_1(x, \mathrm{i}\nu) + I_2(x, \mathrm{i}\nu)}{a(x, \mathrm{i}\nu)} \mathrm{e}^{\mathrm{i}\nu y}\, d\nu\ . \quad (8)$$

Proof: Under the Sobolev assumption, the inverse Fourier formula holds at all $y \in \mathbb{R}$, $m_x(y) = \frac{1}{2\pi}\int_{\mathbb{R}} m_x^*(\nu)\mathrm{e}^{\mathrm{i}\nu y}\, d\nu$. Next we show that

$$m_x^*(\nu) = \frac{I_1(x,\mathrm{i}\nu) + I_2(x,\mathrm{i}\nu)}{a(x,\mathrm{i}\nu)} \tag{9}$$

holds for all $\nu \in \mathbb{R}$ and (8) will thus follow.

Using (4), (5) and (2), the Fourier inversion formula gives, for all (x,p) in $\mathbb{R} \times \mathbb{C}_+$,

$$a(x,p) = \exp\left(\lambda \int_0^\infty \mathrm{e}^{-py} k(x,dy)\right) .$$

where the function k is defined in (3). Taking the log in the last display and applying Fubini's theorem, we get

$$\begin{aligned} \log a(x,p) &= \lambda \mathbb{E}\left[\mathrm{e}^{-pY}(x-X)_+\right] \\ &= \lambda \int_0^x \mathbb{E}\left[\mathrm{e}^{-pY}\mathbb{1}(X \le u)\right] du , \end{aligned} \tag{10}$$

where $u_+ \stackrel{\text{def}}{=} \max(u,0)$.

The rest of the proof consists of two steps: compute the partial derivative $\partial \log a(x,p)/\partial x$ and identify this derivative with the integrand of the RHS of (10). Observe that since $|\mathrm{i}\omega\Psi(\omega,p)|$ is of order ω^{-1} as ω tends to infinity, differentiating the integrand in (5) leads to a non necessarily convergent integral. As observed by [18] in a related problem, it is possible to get rid of this difficulty by finding an explicit expression of the singular part of this function and compute its derivative in the time domain, which yields to the term I_1, while the remainder is differentiated in the frequency domain, which yields to the term I_2. More precisely, using that, for any s in $\mathbb{C}_+^*$, $|\lambda/(s+\lambda)| < 1$, we have, for all $(s,p) \in \mathbb{C}_+^* \times \mathbb{C}_+$,

$$\frac{1}{s+\lambda-\lambda\mathcal{L}P'(s,p)} = \frac{1}{s+\lambda}\sum_{n\ge 0}\left(\frac{\lambda\mathcal{L}P'(s,p)}{s+\lambda}\right)^n . \tag{11}$$

Inserting this in (4), we may write $\Psi(\omega,\mathrm{i}\nu) = \Psi_1(\omega,\mathrm{i}\nu) + \Psi_2(\omega,\mathrm{i}\nu)$, where

$$\Psi_1(\omega,p) \stackrel{\text{def}}{=} \frac{\lambda\mathcal{L}P'(c+\mathrm{i}\omega,p)}{(c+\mathrm{i}\omega+\lambda)^2} , \tag{12}$$

$$\Psi_2(\omega,p) \stackrel{\text{def}}{=} \frac{\lambda\mathcal{L}P'(c+\mathrm{i}\omega,p)}{c+\mathrm{i}\omega+\lambda}\Psi(\omega,p) . \tag{13}$$

Using (5), (12) and (13), we get, for all $p \in \mathbb{C}_+$, $a(x,p) = 1 + a_1(x,p) + a_2(x,p)$, where

$$a_1(x,p) \stackrel{\text{def}}{=} \frac{\mathrm{e}^{(c+\lambda)x}}{2\pi}\int_{-\infty}^{+\infty} \Psi_1(\omega,p)\mathrm{e}^{\mathrm{i}\omega x}\, d\omega$$

$$a_2(x,p) \stackrel{\text{def}}{=} \frac{\mathrm{e}^{(c+\lambda)x}}{2\pi}\int_{-\infty}^{+\infty} \Psi_2(\omega,p)\mathrm{e}^{\mathrm{i}\omega x}\, d\omega$$

Using that $s \mapsto \lambda^2/(s+\lambda)^2$ is the moment generating function of a Gamma distribution with shape parameter 2 and scale λ, we have that Ψ_1 is the product of two moment generating functions and the first inverse Fourier transform appearing in the previous display has an explicit expression in the form of a convolution; more precisely, one obtains, for all $p \in \mathbb{C}_+$,

$$\begin{aligned} a_1(x,p) &= \lambda \iint_{\mathbb{R}_+^2} (x-u)_+ \, \mathrm{e}^{\lambda u - pv} P'(du, dv) \\ &= \lambda \int_0^x I_1(u,\nu)\, du \ , \end{aligned}$$

where the second inequality directly follows from (6). Since $a_2(x,p) = (2\pi)^{-1} \int_{-\infty}^{+\infty} \Psi_2(\omega,p) \mathrm{e}^{(c+\lambda+\mathrm{i}\omega)x} \, d\omega$ and $(c+\lambda+\mathrm{i}\omega)\Psi_2(\omega,p)$ is integrable over $\omega \in \mathbb{R}$, we get that $u \mapsto a_2(u,p)$ is continuously differentiable and, using (7), for all $p \in \mathbb{C}_+$,

$$a_2(x,p) = \lambda \int_0^x I_2(u,\nu)\, du \ .$$

Observing that $\log a(0,p) = 0$, the two last display yields, for all p in $\mathbb{C}_+$,

$$\log a(x,p) = \lambda \int_0^x \frac{I_1(u,p) + I_2(u,p)}{a(u,p)} \, du \ . \tag{14}$$

For any $p \in \mathbb{C}_+$, $I_1(u,p)$ and $I_2(u,p)$ are respectively (at least) right-continuous and continuous, implying that the integrand in (14) is right continuous. So is the integrand in (10) by a dominated convergence argument. Hence these two integrands coincide for all $u \geq 0$ and all $p \in \mathbb{C}_+$. Observe now that, for $p = \mathrm{i}\nu$, $\mathbb{E}\left[\mathrm{e}^{-pY} \mathbb{1}(X \leq u)\right]$ is precisely $m_x^*(\nu)$; hence we have shown (9), which concludes the proof. ∎

C. *Construction of the estimator*

The inversion formula in (8) naturally leads to an estimator of m_x by plugging estimators of a, I_1 and I_2 defined in (5), (6) and (7) respectively. Observe that these quantities depend on the unknown λ and P', which are estimated as follows. Since the idle periods are *i.i.d.* with exponential distribution of parameter λ, a natural estimator is the maximum likelihood estimator, given the idle periods $\{Z_k\}_{1\leq k\leq n}$,

$$\hat{\lambda}_n = \left(\frac{1}{n} \sum_{k=1}^n Z_k \right)^{-1} \ . \tag{15}$$

Let $y \to K(y)$ be a non-negative kernel function which integrates to 1, K^* its Fourier transform, and $h > 0$ a bandwidth parameter. The probability measure P' given a sample $\{(X_k', Y_k'),\ k = 1 \ldots n\}$ is estimated by:

$$\widehat{P'}_n(du, dv) \stackrel{\text{def}}{=} \frac{1}{nh} \sum_{k=1}^n K\left(\frac{v - Y_k'}{h} \right) \delta_{X_k'}(du)\, dv \ , \tag{16}$$

where $\delta_{X'_k}$ denotes the point mass measure at X'_k and h is a bandwidth parameter. It is worth commenting on (16). The standard empirical measure corresponds to letting h tend to 0, so that $h^{-1}K\left((v-Y'_k)/h\right)dv$ tends to $\delta_{Y'_k}(dv)$ in the sense of distributions. Thus $\widehat{P'}_n$ is an estimator of the joint distribution of (X',Y'), regularized along the energy coordinate. It is standard to use a smoothing kernel in order to estimate a density. For instance, $\int_u \widehat{P'}_n(du,dv) = (nh)^{-1}\sum_{k=1}^n K\left((v-Y'_k)/h\right)dv$ yields the standard kernel estimator of the density of Y' with bandwidth h. In fact, h will also act as a bandwidth parameter for our estimator of m defined below in (21) by replacing P' by $\widehat{P'}_n$ in the formula obtained in Proposition 2.1. More precisely, the Laplace transform $\mathcal{L}P'$ is estimated by

$$\begin{aligned}\widehat{\mathcal{L}P'}_n(s,\mathrm{i}\nu) &\stackrel{\text{def}}{=} \iint_{\mathbb{R}_+^2} \mathrm{e}^{-su-\mathrm{i}\nu v}\,\widehat{P'}_n(du,dv)\\ &= \frac{1}{n}\sum_{k=1}^n \mathrm{e}^{-sX'_k-\mathrm{i}\nu Y'_k}K^*(h\nu) \qquad (17)\end{aligned}$$

and the estimators $\widehat{a}_n$ and $\hat{I}_{2,n}$ of a and I_2 respectively are obtained by plugging directly estimators (15) and (17) in (5) and (7):

$$\begin{aligned}&\widehat{a}_n(x,\mathrm{i}\nu) \stackrel{\text{def}}{=} 1+\frac{1}{2\pi}\times\\ &\int_{-\infty}^{+\infty}\frac{\hat{\lambda}_n\widehat{\mathcal{L}P'_n}(c+\mathrm{i}\omega,\mathrm{i}\nu)\mathrm{e}^{(\hat{\lambda}_n+c+\mathrm{i}\omega)x}}{(c+\mathrm{i}\omega+\hat{\lambda}_n)(c+\mathrm{i}\omega+\hat{\lambda}_n-\hat{\lambda}_n\widehat{\mathcal{L}P'_n}(c+\mathrm{i}\omega,\mathrm{i}\nu))}\,d\omega \qquad (18)\\ &\hat{I}_{2,n}(x,\mathrm{i}\nu) \stackrel{\text{def}}{=} \frac{\mathrm{e}^{(c+\widehat{\lambda}_n)x}}{2\pi}\times\\ &\int_{-\infty}^{+\infty}\frac{(\widehat{\mathcal{L}P'}_n(c+\mathrm{i}\omega,\mathrm{i}\nu))^2\mathrm{e}^{\mathrm{i}\omega x}}{(c+\mathrm{i}\omega+\widehat{\lambda}_n)(c+\mathrm{i}\omega+\hat{\lambda}_n-\widehat{\lambda}_n\widehat{\mathcal{L}P'}_n(c+\mathrm{i}\omega,\mathrm{i}\nu))}\,d\omega\ , \qquad (19)\end{aligned}$$

Similarly the estimator $\hat{I}_{1,n}$ of I_1 is defined by replacing λ and P' by their estimates in (6), that is

$$\begin{aligned}\hat{I}_{1,n}(x,\mathrm{i}\nu) &\stackrel{\text{def}}{=} \iint_{\mathbb{R}_+^2} \mathbb{1}_{\{u\le x\}}\mathrm{e}^{\widehat{\lambda}_n u-\mathrm{i}\nu v}\,\widehat{P'}_n(du,dv)\\ &= \frac{1}{n}\sum_{k=1}^n \mathbb{1}_{\{X'_k\le x\}}\mathrm{e}^{\widehat{\lambda}_n X'_k-\mathrm{i}\nu Y'_k}\,K^*(h\nu)\ . \qquad (20)\end{aligned}$$

We finally plug these estimates in (8) and deduce the following estimator for the marginal pdf, which can be computed in practice using a fast Fourier transform (FFT). Details on the implementation can be found for instance in [23] and [24].

$$\begin{aligned}&\widehat{m}(y;c,x,h,n) \stackrel{\text{def}}{=} \frac{1}{2\pi}\times\\ &\qquad\int_{-\infty}^{+\infty}\frac{\hat{I}_{1,n}(x,\mathrm{i}\nu)+\hat{I}_{2,n}(x,\mathrm{i}\nu)}{\widehat{a}_n(x,\mathrm{i}\nu)}\mathrm{e}^{\mathrm{i}\nu y}\,d\nu\ . \qquad (21)\end{aligned}$$

The complete algorithm of pileup correction can thus be detailed as follows :

Step 1) Computation of the lengths of busy and idle periods and energies of the busy periods.

Step 2) Computation of $\hat{\lambda}_n$, $\widehat{\mathcal{L}P'}_n$ and $\hat{I}_{1,n}$ as defined in (15), (17) and (20) (the last two are very similar).

Step 3) Computation of the estimators $\hat{a}_n$ and $\hat{I}_{2,n}$ using a Riemann approximation of the integrals in (18) and (19).

Step 4) Computation of $\widehat{m}(y; c, x, h, n)$ using an FFT algorithm for performing the integral in (21).

It is important to notice that Step 1 of our algorithm is not straightforward. Indeed, standard nuclear physics instrumentation systems only provide the energy associated to busy periods, whereas we both need idle and busy period durations. A possible way to implement it can be found in [25], [26]. Here we assume that we dispose of n independent samples $\{(X'_k, Y'_k),\ 1 \dots n\}$ and $\{Z_k,\ k = 1 \dots n\}$. In the next section we present some numerical methods to choose the tuning parameters of the estimator $\widehat{m}(y; c, x, h, n)$ automatically.

III. DATA-DRIVEN CHOICE OF THE PARAMETERS

Observe that the main difference between the estimator in (21) and the estimator proposed in [19] lies in the kernel regularization, which is done before the Fourier inversion *w.r.t.* the first component in our estimator and after in [19]. Since our estimator is nonlinear, these two ways, though numerically close, are not equivalent. From a theoretical point of view the properties of our estimator can be derived along the same line as in [19], where bounds of the mean integrated squared error are obtained under appropriate Sobolev smoothness assumptions for m and assumptions on the tail behavior of X. Such results are of practical importance, since we often dispose in our application of a large number of samples. Another interest of the theoretical study in [19] for the present estimator is that it gives meaningful explanations about the role of the tuning parameters c, x and h. In short,

1) c is an arbitrary constant used for numerical purposes which does not influence the rate of convergence.
2) x influences the bias as one estimates m_x instead of m. This bias can be controlled if X has light tails.
3) h is a classical bandwidth parameter which influences the bias related to the smoothness of m.

Based on precise bias and variance bounds, choices of x and h are proposed in [19] in order to optimize the rate of convergence. However they depend on the regularity of m and on the tail behavior of X, which are unknown in practice. Here we adopt a more practical point of view, by proposing a data-driven choice of the parameters c, x and h. In particular, the choice of h is motivated by the specific form of our estimator, as discussed below. We also include short discussions about the numerical computation of the integrals appearing in (18), (19) and (21).

a) Choice of c: Due to theoretical properties of the Laplace transform and the Bromwich inversion formula (see e.g. [27]), the choice of c seems of little importance, provided it is positive. In practice this means that as c decreases, the numerical computations of the integrals in (18) and (19) require a finer grid in the Rieman approximation. Nevertheless the variance bounds obtained in [19] indicate that a too large c may increase the variance, but not significantly if it is negligible with respect to λ. Since λ is unknown, a meaningful way for c to be of negligible order is to choose it with same or smaller order than the error made in the estimation of λ. Hence we set $c = \widehat{\lambda_n}/(10\sqrt{n})$, that is, since $\sqrt{n}(\widehat{\lambda_n} - \lambda)/\widehat{\lambda_n}$ converges to a centered unit variance normal random variable, $c/\{10\,|\widehat{\lambda_n} - \lambda|\}$ is bounded in probability.

b) Choice of x: The choice of x can be more problematic. If this parameter is chosen too small, a deterministic bias is introduced which cannot be compensated by a better numerical approximation. On the other hand, if x is chosen too large, the terms of variance developed in [19] cannot be controlled. To handle this problem, we propose to estimate the cumulative distribution function (cdf) associated to X using observations of X' (see *e.g.*[28]) and choose x as the corresponding 90% quantile. This method allows to take the most representative busy periods, while keeping x small enough to ensure the control of the variance terms of [19]. In the practical application that we have considered, X has an upper bound, say $\bar{x}$, so taking $x \geq \bar{x}$ implies $m_x = m$ and the bias related to the choice of x vanishes. However, even in this case, the method that we proposed is useful to estimate $\bar{x}$, since the 90%-quantile of X can be robustly estimated, in contrast to its maximal value; in fact, the maximal value of X'_k, $k = 1, \dots, n$ tends to ∞ as $n \to \infty$, thus giving a bad estimate of $\bar{x}$, as it will be shown in the next section.

c) Choice of h: The bandwidth parameter h is used for regularization. The choice of this parameter is more delicate than the previous ones because one has to adapt to the smoothness of the density m of Y to be estimated. In contrast the choice of x is related only to the tail properties of X, not on properties of m. This parameter is standard in density estimation : a too small setting of h induces a well-known Gibbs phenomenon around the energy peaks, whereas a too large h is susceptible to mask small peaks and deteriorates the resolution. Although numerous methods have been developed for automatic selection of the bandwidth parameter in density estimation (one can refer to [29] for a nearly exhaustive comparison of their performances), such methods cannot be applied directly as direct observations of Y are not accessible. However contrary to the post-regularization used in [19] our regularization applies on the empirical measure of P', see (16). Therefore, a way to choose automatically h is to use a bandwidth selector (say, by minimizing a cross validation criterion) on the observations $\{Y'_k,\ k = 1, \ldots, n\}$ as one would do for estimating the density of Y' and then plug this value in the estimator (21) of m. It must be noticed that a data-driven bandwidth selection will generally optimize the bias/variance trade-off on the whole spectrum. If one is more interested in optimizing the resolution of the peaks, a lower value of h should be chosen.

d) Discretization of the integral in (21): Suppose that the density m is estimated on the interval $[0; y_{max}]$ on a number of points characterized by the sharpness of a subdivision Δy. In practice, the integral in (21) uses a Riemann approximation (computed via an FFT algorithm), so we need further heuristics concerning the choices of:

- the supremum according to the second dimension ν_{max} and
- the sharpness of the subdivision used for the integral approximation in (21), denoted by $\Delta\nu$.

The maximum energy y_{max} and the the subdivision size Δy are used to set the values of ν_{max} and $\Delta\nu$ corresponding to the Shannon's theorem for discrete sampling of band-limited functions. By doing so, we of course introduce some aliasing which can pollute the estimates when ν_{max} is too small and $\Delta\nu$ too large. More precisely, we choose in practice

$$\nu_{max} = \frac{2\pi}{\Delta y} \text{ and } \Delta\nu = \frac{2\pi}{y_{max}} .$$

e) Discretization of the integrals in (18) and (19): We need to control the numerical error in the approximation of the integrals in (18) and (19). Hence, we provide heuristics concerning the choice of

- the supremum according to the first component ω_{max},
- the sharpness of the subdivision $\Delta\omega$ used for the Riemann approximation of the integrals.

Observe that, using that $|\widehat{\mathcal{L}P'}_n| \leq 1$ in (18) and (19), the corresponding integrand and their derivative can be easily bounded by integrable functions which do not depend on ν. Therefore, once a threshold ε is fixed by the user, it is straightforward to choose ω_{max} and $\Delta\omega$ in order to impose a Riemann approximation error below ε. In practical applications, we fixed ε such that the deterministic error made in the integral approximation is negligible with respect to the estimated pdf in the energies of interest.

IV. Results on simulations

We present in this section results on the parameters automatic selection methods discussed in Section III on simulations. For $\lambda = 0.04$ (which corresponds to a physical counting rate of 400000 photons per second), we generate a sample of durations and energies of the individual pulses according to the following

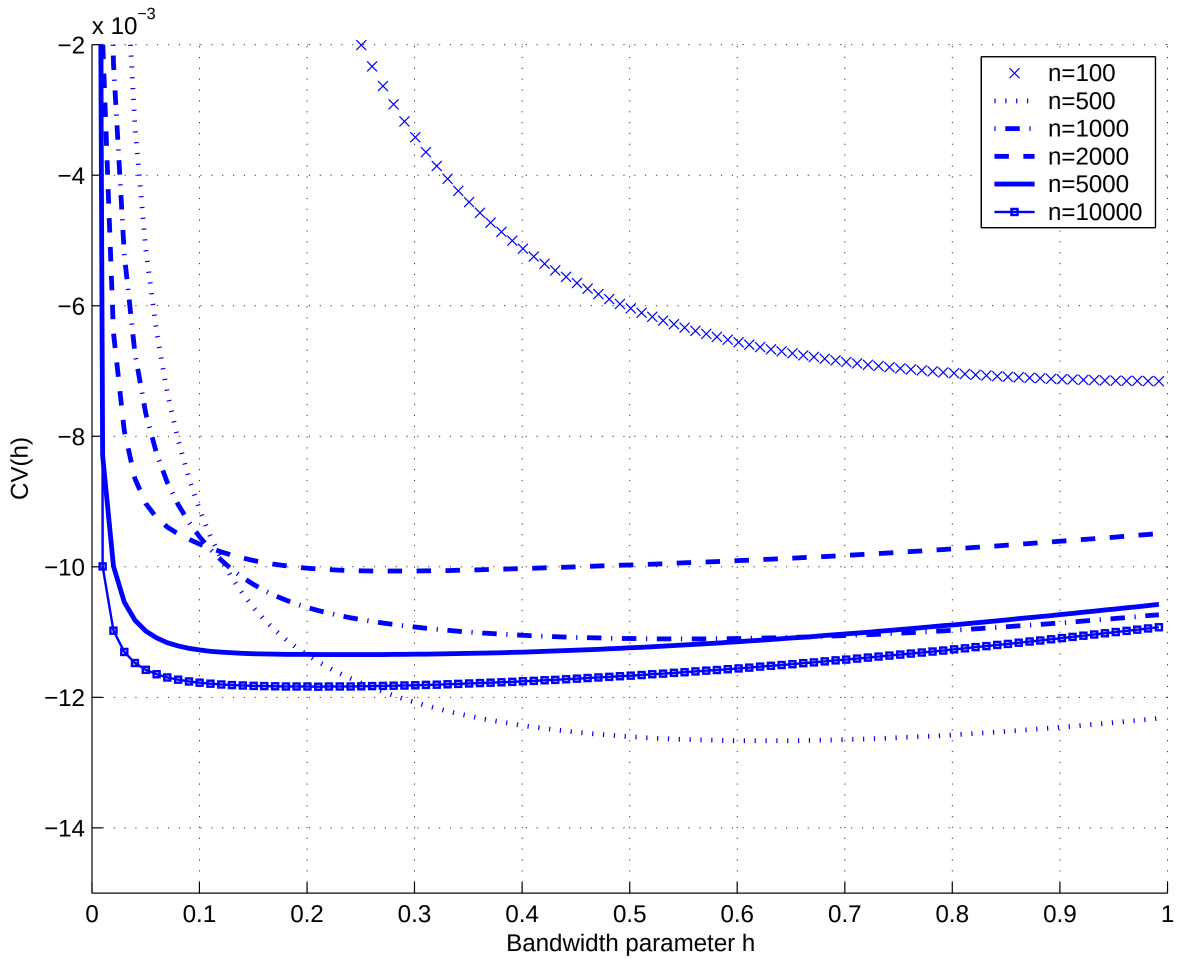


Fig. 3. The cross-validation criterion $CV(h)$, with different sample sizes n.

distribution:

$$f(x,y) = m(y) \times f_{X|Y}(x|y) \;, \tag{22}$$

where m is represented by the black plot of Figure 2 and the conditional distribution $f_{X|Y}(\cdot|y)$ is a truncated Gamma distribution $\Gamma(\alpha,\beta,T)$ with $\alpha = 2 + y/1024$, $\beta = 1$ and $T = 4 + y/2048$. We draw samples in order to dispose of $n = 10000$ observations $\{(X'_k, Y'_k),\ k = 1\ldots n\}$ using an Acceptation-Reject procedure as described in Chapter 2 of [30].

Denote by g the pdf associated to Y', and by $\hat{g}$ a kernel estimate of g, that is:

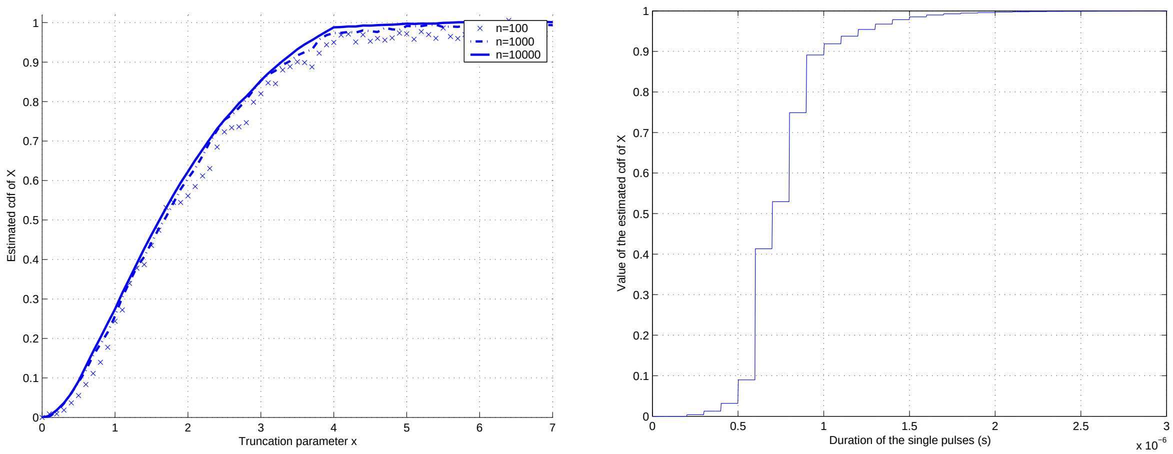
$$\hat{g}(y) = \frac{1}{nh}\sum_{k=1}^{n} K\left(\frac{y - Y'_k}{h}\right) \;,$$

and define $CV(h)$ as the usual cross-validation criterion to be minimized for the data-driven choice of

h:

$$CV(h) \stackrel{\text{def}}{=} \int \hat{g}^2 + \frac{2}{n(n-1)h} \sum_{k=1}^{n} \sum_{k' \neq k} K\left(\frac{Y'_{k'} - Y'_k}{h}\right)$$

which is an unbiased estimate of the mean integrated error up to an additive constant not depending on h, namely $\mathbb{E}\left[CV(h)\right] = \mathbb{E}\left[\int \hat{g}^2 + 2 \int g\,\hat{g}\right] = \mathbb{E}\left[\int (\hat{g} - g)^2\right] + \int g^2$. Figure 3 shows unbiased estimates of the cross-validation function, for different values of n. Figure 4(a) shows the estimate of the cdf of



(a) estimates of the cdf of X for simulations, with different number of samples n

(b) Estimate of the cdf of X for real data

Fig. 4. Estimates of the cdf of X for simulations and real data

X, obtained from the samples $\{X'_k,\ k = 1 \dots n\}$, using the previously described methodology and for different values of n up to $n = 10000$. Finally, Figure 5 represents the estimate of m using the pileup correction method, with $c = 0.0001$ and the parameters x and h being chosen using the data-driven parameter selectors previously described. It shows good similarity between the estimator and the true pdf, which validates this approach.

V. Applications and discussion

It is important to notice that the proposed method relies on the knowledge of the durations and energies associated to the busy periods. Therefore, a good signal-noise ratio is necessary to compute our samples before estimating m, since it is required to detect the beginning and the end of each busy period as precisely as possible. Many methods have been proposed to detect precisely the beginning of busy periods, based on the matched filter framework, as in [31]. However, more recent advances (see [32] and

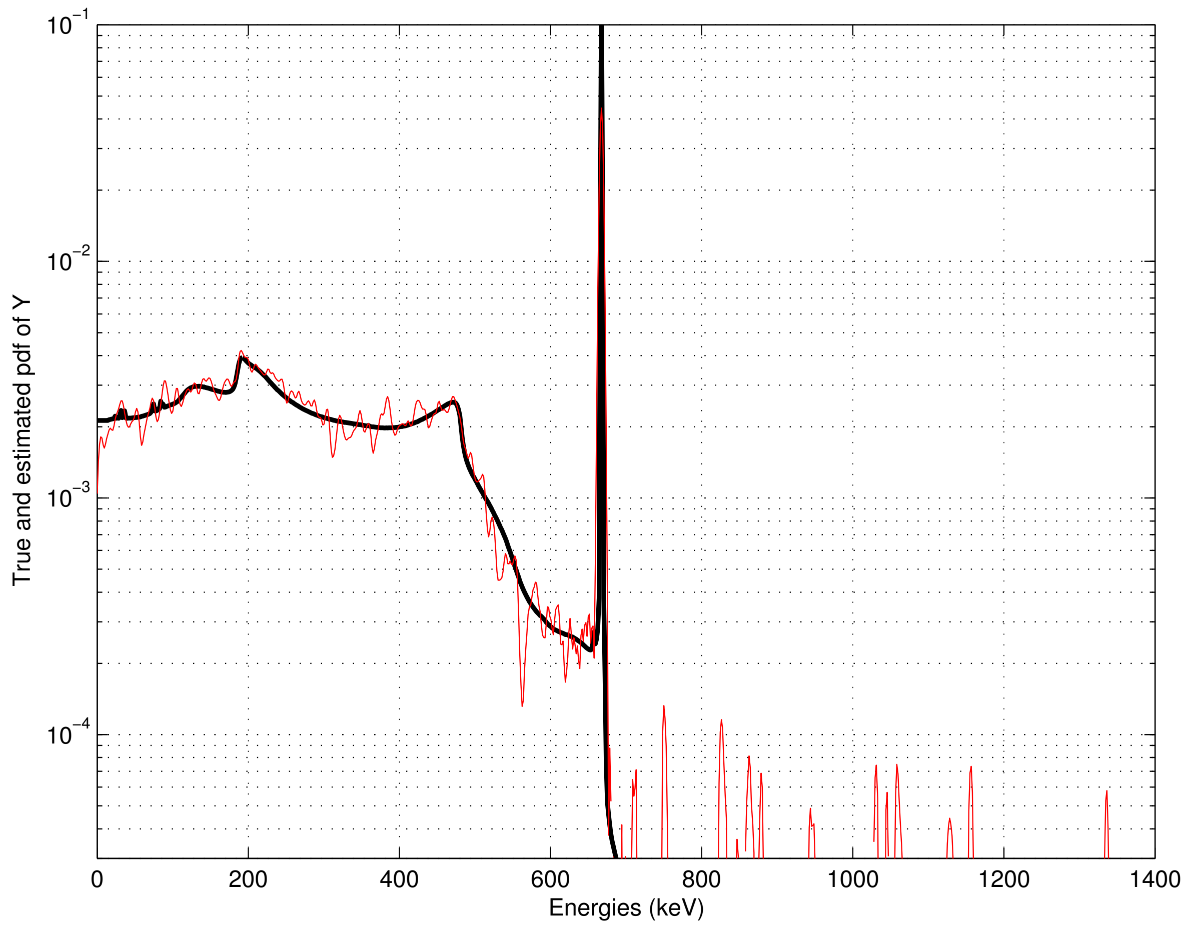


Fig. 5. Comparison between m (black curve) and its estimate (red curve).

[33]) allow to deal with higher counting rates. The couples $\{(X'_k, Y'_k), k = 1 \dots N\}$ were obtained using the instrumentation described in [32], and we refer to this contribution for further details on this issue.

A. Experimental setup

We now observe $n = 10^6$ cycles obtained using the ADONIS system (see [34], [35], [32] and [26]), from a mixture of Cesium 137 (which is ideally characterized by a single energy peak at 662 keV), Cesium 134 (which has some representative energy peaks at and 604 and 795 keV) and Americium (whose representative peaks are in 16, 55, 721 and 759 keV), and we wish to retrieve its associated energy spectrum. The energy spectrum before pileup correction is displayed in Figure 6(a). Given these observations, we apply the algorithm described in Section II. The estimated intensity is $\hat{\lambda}_n \approx 210000$ photons per second. We set c and h as described in the previous section. The maximum length of the busy periods we observed is $6.1\,\mu s$, so choosing x at least to this value means that all the observations X'_k are used for computing $\hat{I}_{1,n}$ in (20). We present results for $x = 6.1\,\mu s$, $x = 1\,\mu s$ and $x = 0.7\,\mu s$. The two last values approximately corresponds respectively to the 90%-quantile and the median associated to the estimated cdf of X, using the methodology described in [28], as it can be seen in Figure 4(b). We observe that the estimate of the cdf of X is quantized in the case of real data; this is not due to our choice of estimator (as it gives a continuous estimate in the case of simulations in Figure 4(a)), but rather to the fact that our data are provided by a digital instrumentation (see paragraph V-E for in-depth discussion on this aspect). In Figure 6, we plot the results obtained by our pileup correction method in these various situations and discuss the results hereafter.

B. *Results for $x = 1\,\mu s$*

This choice of x corresponds to the 90%-quantile of the cdf of X (see Figure 4(b)), so that m_x remains close to m. Figure 6(a) shows the spectrum obtained by our pileup correction method (red curve) and the energy spectrum obtained without any correction (blue curve). We can see that the Compton continuum and the fake peaks of the spectrum have been corrected.

We then focus on the zone of this energy spectrum which is approximately in $[600\,keV\ ;\ 800\,keV]$, which is enlarged in Figure 7. Although the resolution is affected by the choice of the bandwidth h for the corrected spectrum, several peaks appear clearly, among them, the 662 keV peak of the Cesium 137, the 604 keV and 795 keV peaks of the Cesium 134, and 725 keV peak and 759 keV of the Americium 241, whose main representative peaks at 16 keV and 59 keV are hidden in the Compton continuum.

Finally, if we look on the zone of this energy spectrum which is in $[800\,keV\ ;\ 1\,MeV]$, we observe new peaks after pileup correction.

Comparing this figure to the tables given in the reference database JEF 2.2, we remark that observed peaks show strong similarities with many energies described in our tables. For example, we retrieve by applying our algorithm the presence of Cesium 132, Cesium 130 and Cesium 129. However, due to the small dataset used in this experiment, it is hazardous to conclude on the nature of these peaks, real or not.

C. *Results for $x = 6.1\,\mu s$ and $x = 0.7\,\mu s$*

With those choices of x, we either set x to $\max_k X'_k$, that is the smallest value such that the indicator in (20) is always one, either to the estimated median of X. Our results are presented in Figures 6(b) and 6(c). As mentioned previously, a too large x causes high variability in our estimator, as implied by the theoretical bounds established in [19]. Comparing the blue curve in Figure 6(a) and Figure 6(b), it appears that the observed spectrum and the estimated spectrum have similar aspects except for a higher variability in the latter case, since the terms of variance as described in [19] are less likely to be controlled.

Comparing Figures 6(a), 6(b) and 6(c) shows that the parameter x is a sensitive one for the case of real data. in particular in the tail behavior of X. It suggests a strong dependency between X and Y. Indeed, we recall that in the case where X and Y are independent, m_x is equal to m up to a multiplicative constant $\mathbb{P}(X \leq x)$.

D. Validation

In order to validate our estimator on real data (that is, to check if the newly observed spikes are not only due to variability), it may be useful to compare our results with what could be obtained using a pileup rejector method. Although many rejectors based on the time signal have been proposed, they are not appliable here, since we only dispose on the information on durations and total energies busy periods. In our context, a simple rejection method is obtained by thresholding the durations of busy periods adequately, so that a large proportion of the remaining observations correspond to single pulse busy periods.

The rejection method thus consists in estimating Y' conditionnally on $X' \leq t$, where t is the threshold. For t small enough, the proportion of single pulse busy periods is close to one, and we obtain an estimator of Y conditonally on $X \leq t$. Hence the choice of t is very similar to the choice of x in Section III, where the distribution of Y is also approximated by conditionning on $X \leq t$. In our experiments, we thus took $t = 1\,\mu s$. Now, contrary to the correction estimator, the observations having durations larger than $1\mu s$ are not used. We show the results of the rejection method with $n = 10^7$ samples instead of 10^6 as previoulsy used for the correction method. Indeed, applying our method of rejection on only 10^6 samples gave poor results. The obtained energy spectrum is plotted in Figure 8(a), and a zoom on the region $[600\ keV; 800\ keV]$ is given in Figure 8(b).

By examining the spectrum after pileup rejection, it appears that the spikes observed in the area between 600 and 800 keV are confirmed by the rejection method, which validates the proposed approach. We conclude this section by stressing that rejection methods, although numerically simpler, are not statistically optimal as they do not use all the information in the data. From the practical point of view, any method of rejection based on duration would automatically reject high energy busy periods, whether they correspond to a real electrical pulse or to a pileup, thus it has to be done carefully. Note moreover than a rejection of pileups would result in a bias in the estimation of the activity of the source λ.

E. Discussion

Considering the reconstructed energy spectrum, the most important thing to see is that the piled-up Compton continuum is well corrected, allowing to see isotopes of the Cesium and Americium that do not appear in the initial spectrum because of the piled-up Compton continuum. We can therefore see more clearly that our radioactive element is a mixture of Cesium 137, Cesium 134 and Americium 241. Observe that improvement was obtained using only 10^6 samples, which is very little data in that framework. Theoretical results of [19] guarantee better results with a more important (and more often encountered in spectrometry) dataset.

Moreover, in the corrected energy spectrum, we now distinguish peaks of other isotopes of the Cesium, which probably are Cesium 132,130 and 129 and were hidden by the piled-up Compton continuum. More precisely, using the JEF 2.2 database for identifying radionuclides, we retrieve some peaks of Cesium 132 (667 keV), Cesium 130 (895 keV) and Cesium 129 (864, 906 and 946 keV). This makes sense physically, since a source of a radioactive element always contain traces of its isotopes, which cannot be separated chemically. Recall nevertheless that due to the small dataset used, it may be hazardous to conclude on these peaks without further investigation. Moreover, a few peaks do not correspond to radioactive elements, or should be subject to further look in the JEF 2.2 database. Nevertheless, our results seem very promising.

We now comment some limitations of this algorithm. One comes from the additive measurement noise, that is not taken into account in our modeling. Indeed, due to the sampling period, we do not observe directly the k-th pileup (X'_k, Y'_k), but rather $(X''_k, Y''_k) = (X'_k, Y'_k) + (\epsilon_k, \eta_k)$, where (ϵ_k, η_k) is the error due to the discretization of the signal. If we assume that this additive error has a probability measure G and is independent of $\{(X'_k, Y'_k),\ k = 1..n\}$, we rather dispose of observations with associated probability measure $P' * G$, thus we should correct the pileup correction formula as follows:

$$\int_0^{+\infty} \mathrm{e}^{-(s+\lambda)u} \left[\exp\left(\lambda \int_0^{\infty} \mathrm{e}^{-pv} k(u,v)\, dv\right) - 1\right] du$$
$$= \frac{\lambda(\mathcal{L}P'(s,p)/\mathcal{L}G(s,p))}{s+\lambda} \frac{1}{s+\lambda-\lambda(\mathcal{L}P'(s,p)/\mathcal{L}G(s,p))}\ .$$

As mentioned above, the ADONIS system is a digital instrumentation, and therefore the sampling of the data can introduce a bias in the measurement of X'. Apart from the error due the discretization in the inversion formula, which has already been exposed, this also leads to two other problems. First, it introduces a quantization of the estimate of the cdf of X, which can be problematic for the choice of x. Moreover, we remark experimentally that the sampling period is only ten times smaller than the mean length of a *busy* period. Consequently, the beginning of some *busy* periods may be missed, and it could indicate that the considered *busy* periods of high length are in fact clusters of *busy* periods, whose beginning or end were not detected due to the sampling.

VI. Conclusion

In this paper we investigated the problem of correcting the pileup phenomenon in γ spectrometry. Using methods inspired from the ones applied for nonparametric density estimation in the deconvolution framework, we proposed an algorithm which gives very promising results on real data.

As mentioned in the discussion, the use of HPGe detectors which limits the problem of the additive noise, and guarantees a good signal-noise ratio, is crucial in our approach. Also, to correct the additive error that comes from the discretization, the application of a density deconvolution technique such as a wavelet-based estimation of P' (see e.g. [36]) before applying our algorithm should improve the results significantly, and could extend the use of this techniques to other types of detectors. Another extension of the pileup correction method is its adaptation to discrete-time signals, in order to be used more efficiently on digital gamma spectrometers. These considerations will be developed in future contributions.

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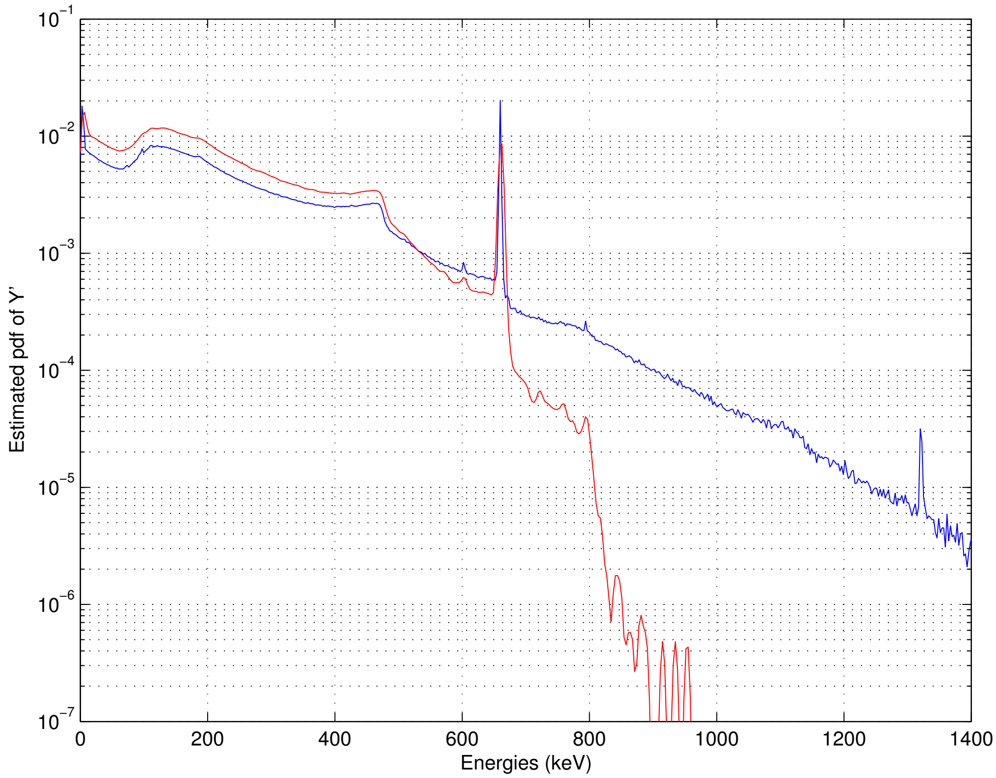
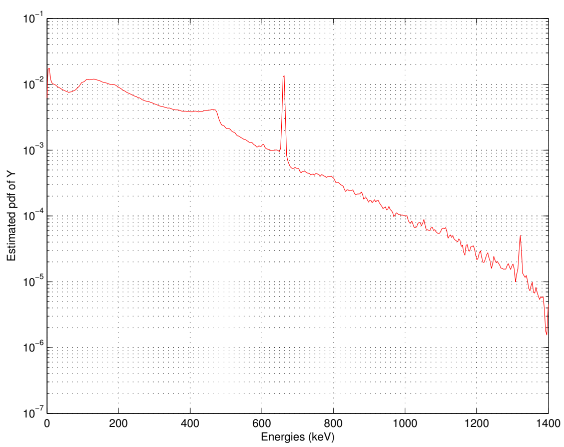
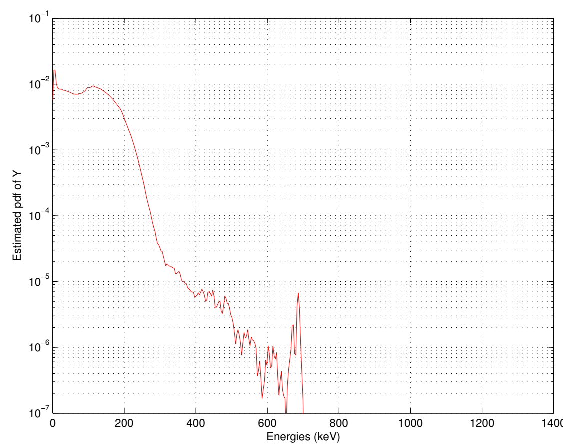


(a) Observed energy spectrum (blue) and corrected estimate for $x = 1\,\mu s$ (red)



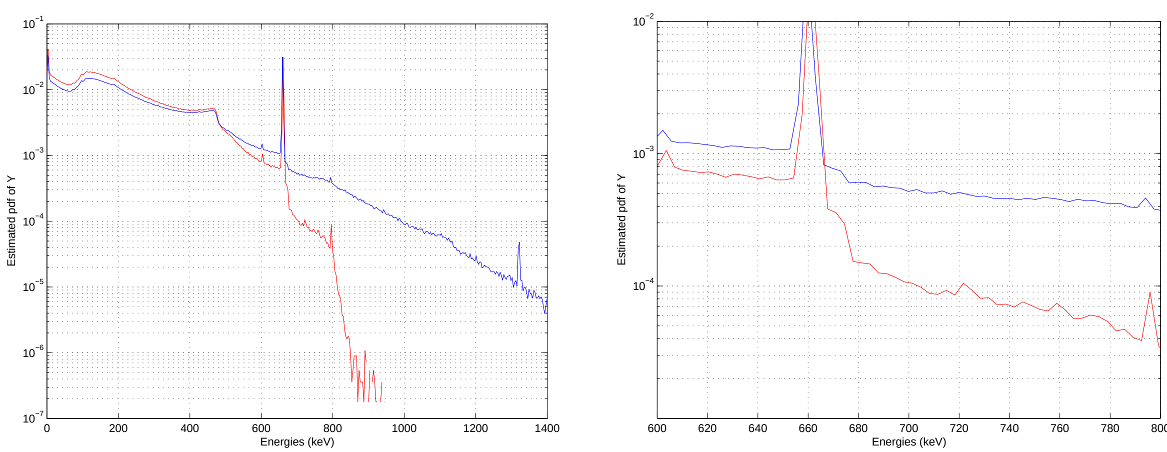
(b) Energy spectrum after pileup correction for $x = 6.1\,\mu s$



(c) Energy spectrum after pileup correction for $x = 0.7\,\mu s$

Fig. 6. Results on real data for different values of x.

Fig. 7. Energy spectrum after pileup correction for $x = 1\,\mu s$ — Enlarged zone $[600\,keV\ ;\ 800\,keV]$.



(a) Energy spectrum (blue curve) and energy spectrum after pileup rejection on 10^7 samples.

(b) Enlarged zone between 600 keV and 800 keV.

Fig. 8. Results of a pileup rejection method for validation of the spikes